1. Problem: simulation labels have unequal value

- A learned surrogate predicts secure generator dispatch.
- The grid changes: line 8 is removed from a six-bus network.
- Target-topology SCED labels have a limited budget.

Question: which operating conditions should we simulate so that each new label improves the downstream decision?

Structural shift
Source: all 10 lines

Target: **line 8 removed**

Matched diagnostic:
+0.077 MW MAE
+0.55 mean regret

2. Gap: prediction error is not operational value

- Coverage finds unusual inputs; uncertainty finds model disagreement.
- Neither explicitly asks whether an error matters for network constraints.
- Dispatch MAE can rank models differently from repair burden and cost regret.

Testable idea

Weight uncertainty by predicted proximity to a base-case or contingency line limit—without observing the hidden SCED label.

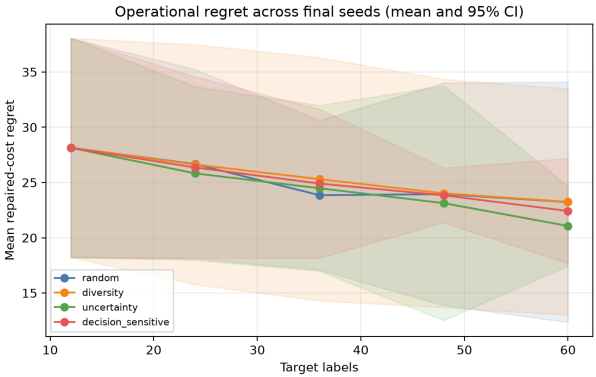
3. Method: one fair, leakage-controlled loop

220 source labels $\rightarrow$ 5-MLP ensemble $\rightarrow$ target pool

Random Diversity Uncertainty $\tilde{u}(1+\tilde{c})$

- 160 candidates; 12 shared initial labels; four batches of 12.
- Same model, seeds, repair, label budgets, and untouched 80-case test set.
- Evaluate raw security feasibility, ℓ_1 repair, and repaired-cost regret.

4. Result: uncertainty wins; weighting can hurt



At 60 labels (three-seed means):

Method	MAE	Regret
Random	1.582	23.23
Diversity	1.504	23.24
Uncertainty	**1.387**	**21.06**
Weighted	1.425	22.42

The decision-sensitive heuristic does not beat uncertainty; one case is worse by 54.253 regret units.

5. PhD extension: from heuristic to value of information

1. Calibrate uncertainty and operational criticality on validation data.
2. Estimate expected repair or regret reduction before querying a label.
3. Scale to multiple outages, larger grids, AC-OPF, and dynamic simulators.
4. Use real measurement correlations to construct credible scenarios.

Take-away

Decision-aware acquisition is worth testing, but it succeeds only when its proxy for downstream value is reliable.